In [24]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import string
import seaborn as sns
import plotly as px
```

In [25]:
```
df = pd.read_excel(r"Products.xlsx")
df.columns = df.columns.str.strip()
df.columns = df.columns.str.lower().str.replace(" ", "_")
df = df.apply(lambda col: col.map(lambda x: x.strip() if isinstance(x, str) el
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 53 entries, 0 to 52
Data columns (total 16 columns):
 #   Column                              Non-Null Count  Dtype
---  ------                              --------------  -----
 0   name                                53 non-null     object
 1   state                               53 non-null     object
 2   location                            53 non-null     object
 3   land_holdings                       6 non-null      object
 4   owner                               53 non-null     object
 5   crops                               53 non-null     object
 6   landform_type                       42 non-null     object
 7   climatic_conditions                 18 non-null     object
 8   innovation_and_practice_uses        52 non-null     object
 9   processing_type                     53 non-null     object
 10  result                              53 non-null     object
 11  profit_increase_or_revenue_growth   53 non-null     object
 12  employment_generation               53 non-null     object
 13  applicability_in_chhattisgarh       53 non-null     object
 14  key_learning                        53 non-null     object
 15  remark                              53 non-null     object
dtypes: object(16)
memory usage: 6.8+ KB
```

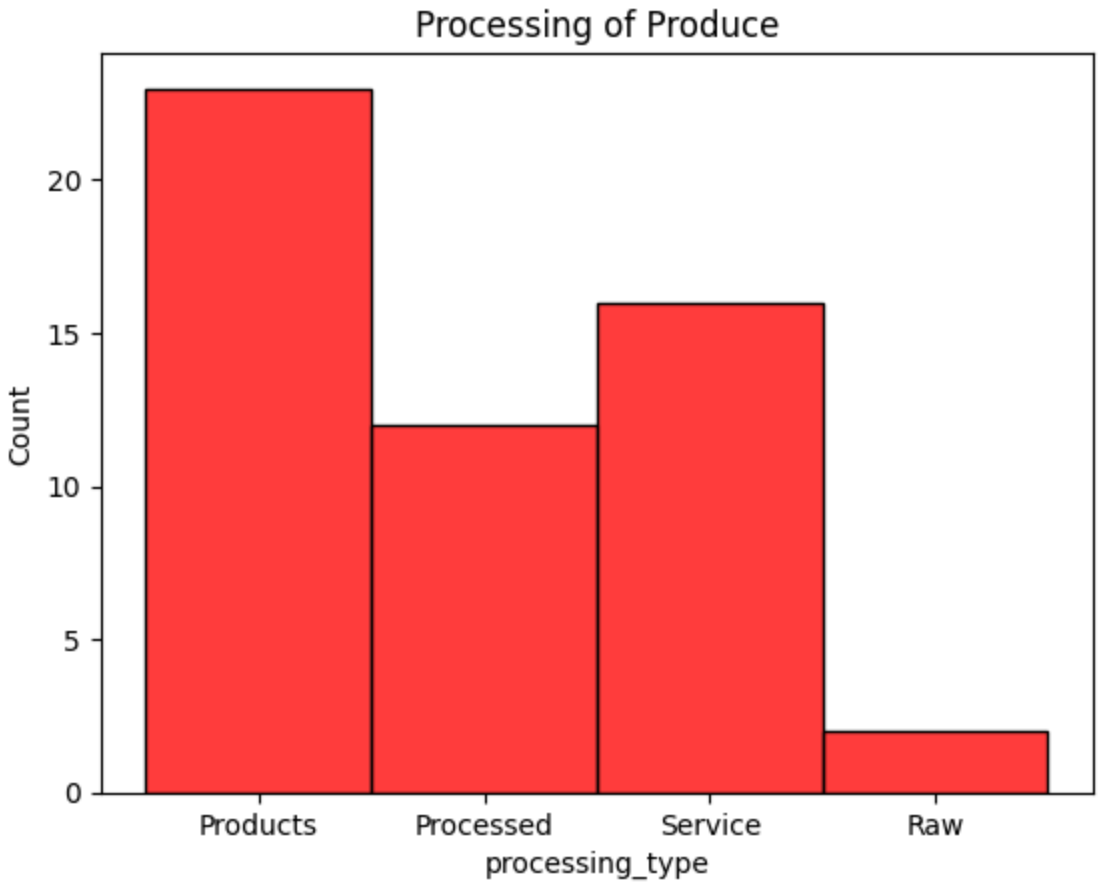
In [26]:
```
df=df.drop(columns=['land_holdings'])
df['crops']=df['crops'].str.lower()
```

In [27]:
```
df['processing_type'].value_counts()
```

Out[27]:
```
processing_type
Products     23
Service      16
Processed    12
Raw           2
Name: count, dtype: int64
```

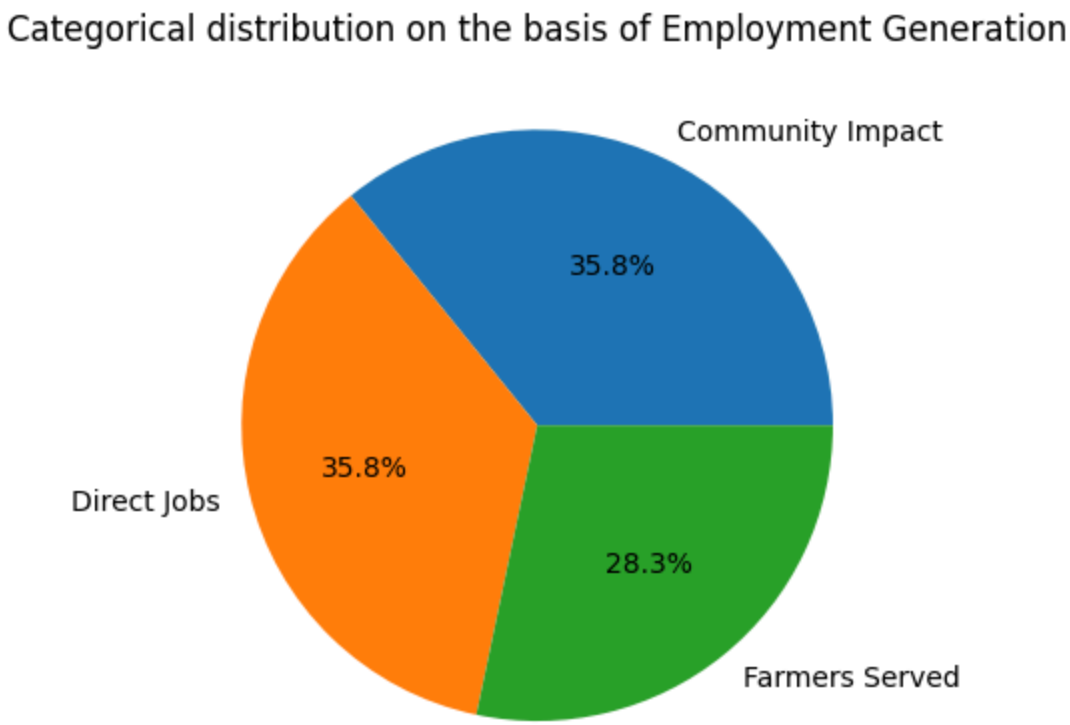
In [28]:
```
sns.histplot(data=df, x='processing_type', color='red')
plt.title("Processing of Produce")
```

Out[28]:
```
Text(0.5, 1.0, 'Processing of Produce')
```



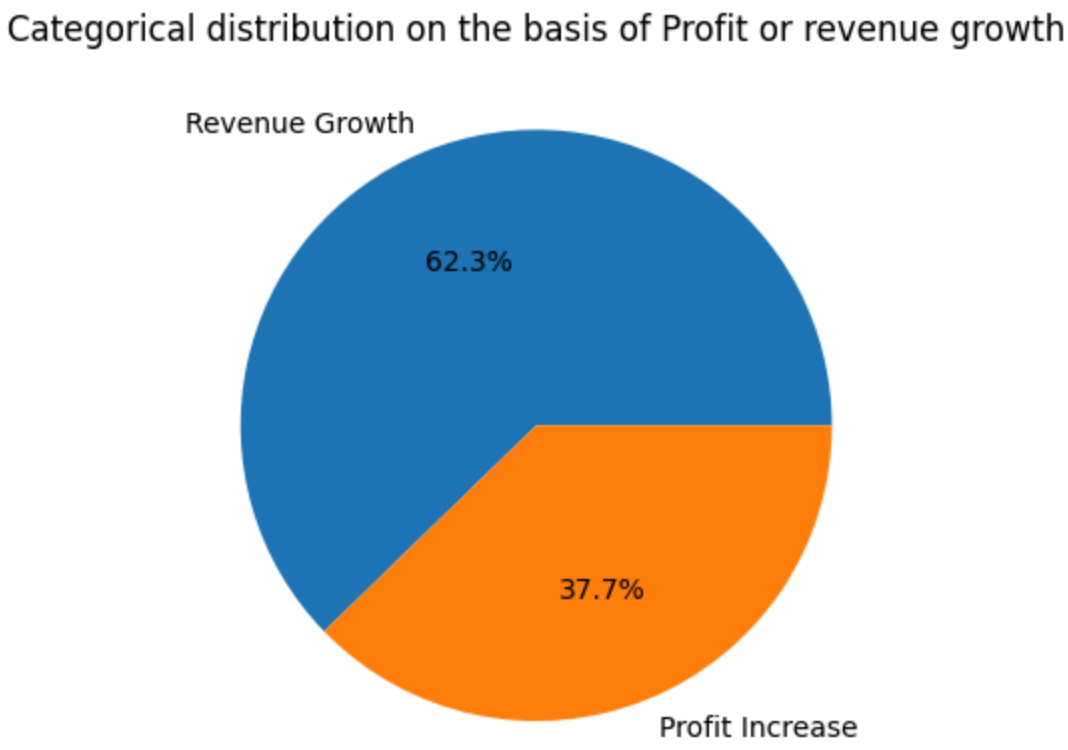
In [29]:
```
plt.pie(df['employment_generation'].value_counts(),labels=['Community Impact',
plt.title("Categorical distribution on the basis of Employment Generation")
df['employment_generation'].value_counts()
```

Out[29]:
```
employment_generation
Community Impact    19
Direct Jobs         19
Farmers served      15
Name: count, dtype: int64
```



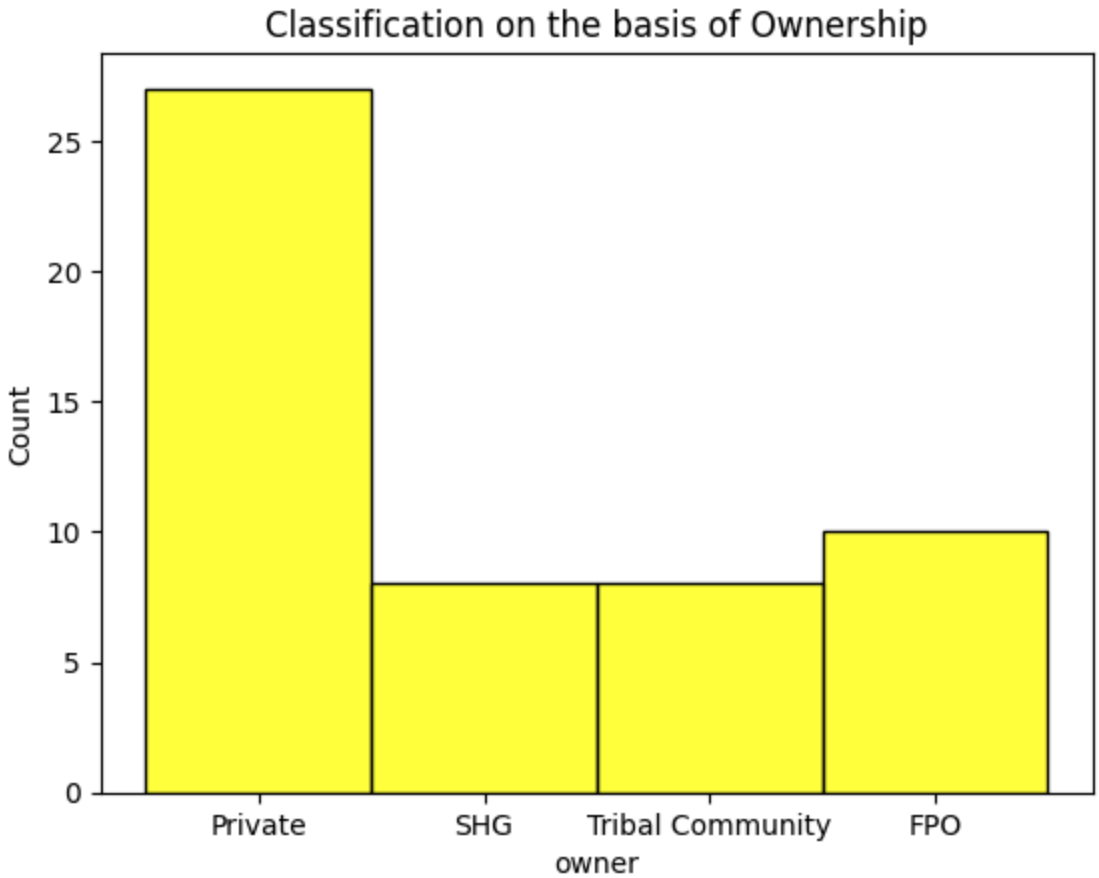
In [30]:
```
plt.pie(df['profit_increase_or_revenue_growth'].value_counts(),labels=['Revenu
plt.title("Categorical distribution on the basis of Profit or revenue growth")
df['profit_increase_or_revenue_growth'].value_counts()
```

Out[30]:
```
profit_increase_or_revenue_growth
Revenue Growth     33
Profit Increase    20
Name: count, dtype: int64
```



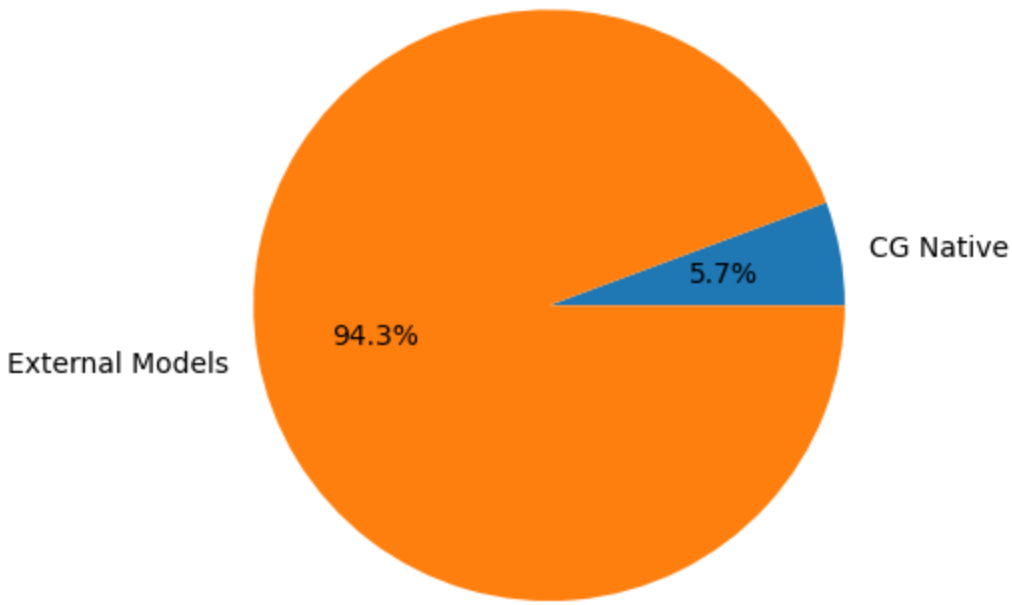
In [31]:
```
sns.histplot(data=df,x=df['owner'],color='yellow')
plt.title("Classification on the basis of Ownership")
```

Out[31]:
```
Text(0.5, 1.0, 'Classification on the basis of Ownership')
```

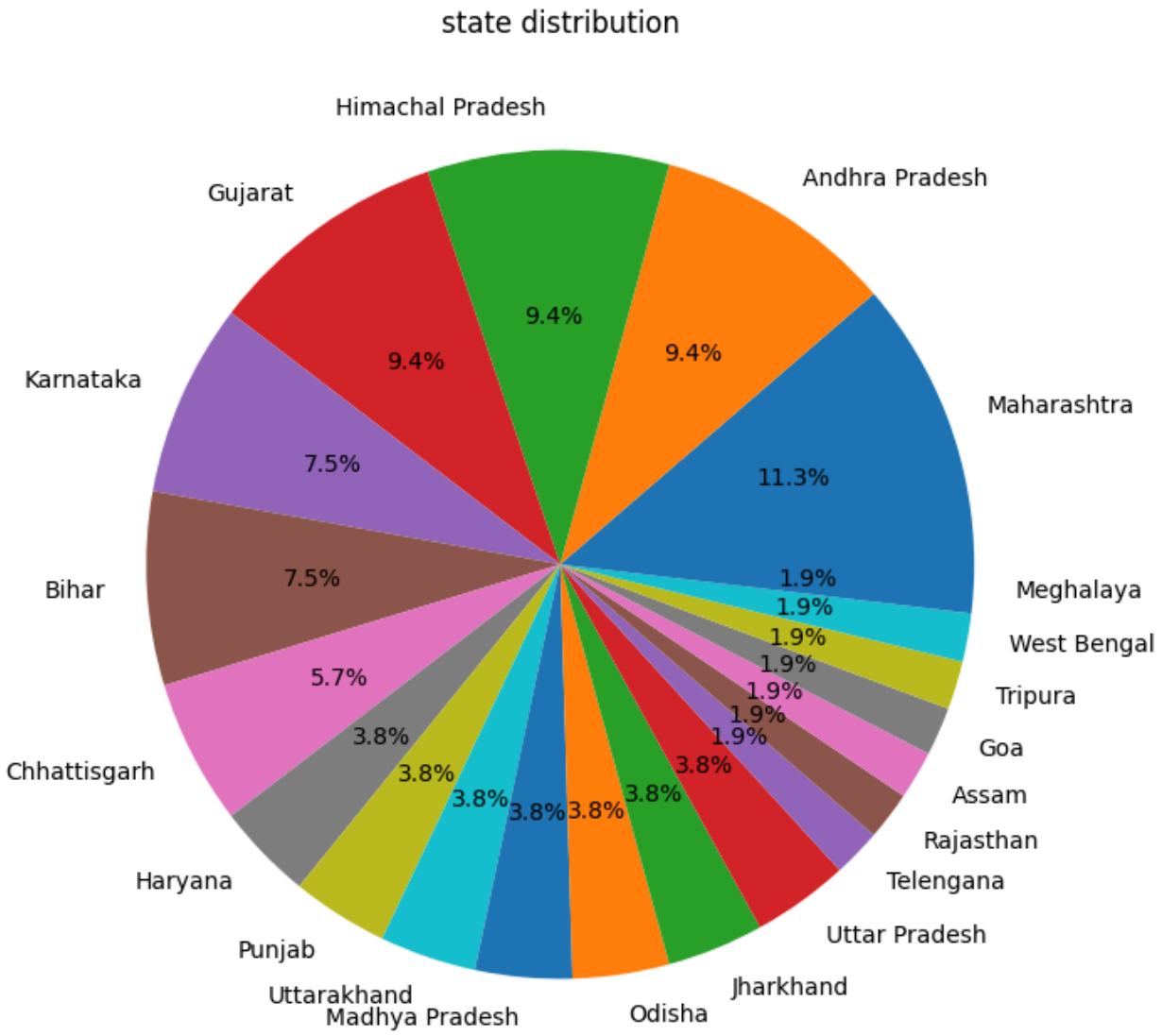


In [32]:
```
# Pie Chart - Native vs External Models
cg_native = df[df['state'] == 'Chhattisgarh']
external = df[df['state'] != 'Chhattisgarh']
pd.Series({'CG Native': len(cg_native), 'External Models': len(external)}).plo
```

Out[32]: <Axes: >



In [33]:
```
counts = df['state'].value_counts()

plt.figure(figsize=(8,8))
plt.pie(
    counts.values,                # sizes
    labels=counts.index,          # labels auto detected
    autopct='%1.1f%%'
)

plt.title("state distribution")
plt.show()
df["state"].value_counts()
```


state distribution
Himachal Pradesh
Andhra Pradesh
Gujarat
Karnataka
Maharashtra
Bihar
Meghalaya
West Bengal
Tripura
Goa
Assam
Rajasthan
Telengana
Uttar Pradesh
Jharkhand
Odisha
Madhya Pradesh
Uttarakhand
Punjab
Haryana
Chhattisgarh
9.4%
9.4%
9.4%
11.3%
7.5%
7.5%
5.7%
1.9%
1.9%
1.9%
1.9%
1.9%
1.9%
1.9%
3.8%
3.8%
3.8%
3.8%
3.8%
3.8%
3.8%

Out[33]:
```
state
Maharashtra         6
Andhra Pradesh      5
Himachal Pradesh    5
Gujarat             5
Karnataka           4
Bihar               4
Chhattisgarh        3
Haryana             2
Punjab              2
Uttarakhand         2
Madhya Pradesh      2
Odisha              2
Jharkhand           2
Uttar Pradesh       2
Telengana           1
Rajasthan           1
Assam               1
Goa                 1
Tripura             1
West Bengal         1
Meghalaya           1
Name: count, dtype: int64
```

In [34]:
```
df[df['state']=='Chhattisgarh']
```

Out[34]:

	name	state	location	owner	crops	landform_type	climatic
10	15 SHGs	Chhattisgarh	Durg	SHG	mushroom (oyster)	Plains	
11	Tribal FPO	Chhattisgarh	Rajnandgaon	FPO	tribal farming	Forest	
50	Millet Café (Raigarh SHG)	Chhattisgarh	Raigarh	SHG	ragi, kodo, and kutki (value added)	NaN	

Forming Categories of the crops

In [35]:
```
product = []
for i in range(0,df.shape[0]):
    s = df.loc[i,'crops']
    s = s.replace("and",'')
    s = s.translate(str.maketrans('', '', string.punctuation))
    words = s.lower().split()
```

```
    product.extend(words)
```

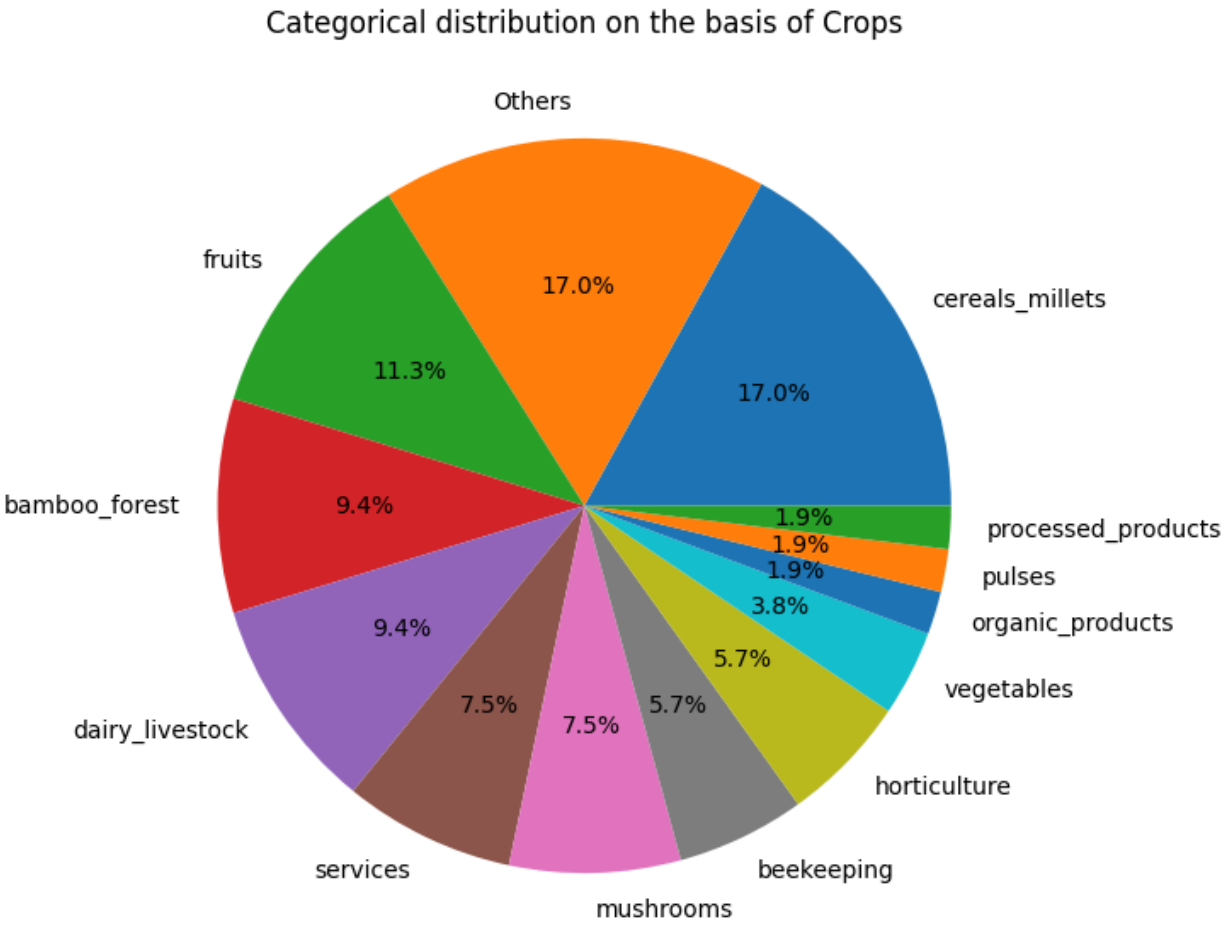
In [36]:
```
product=list(set(product))
stopwords=['green','them','ifs','veggies','tribal','input','processing','bred'
product = [w for w in product if w not in stopwords]
```

In [37]:
```
crop_categories = {

    # Cereals & Millets
    "cereals_millets": [
        "millet", "jowar", "bajra", "foxtail",
        "ragi", "kodo", "kutki",
    ],

    # Pulses / Legumes
    "pulses": [
        "chickpea", "gram", "peas", "pulse", "pulses"
    ],

    # Fruits
    "fruits": [
        "mango", "banana", "papaya", "strawberry",
        "plum", "amla", "jackfruit", "tamarind"
    ],

    # Vegetables
    "vegetables": [
        "tomato", "brinjal", "cucumber", "ginger", "fennel"
    ],

    # Mushroom Farming
    "mushrooms": [
        "mushroom", "oyster"
    ],

    # Dairy & Livestock
    "dairy_livestock": [
        "dairy", "milk", "khoya", "paneer",
        "cows", "livestock", "fodder"
    ],

    # Beekeeping
    "beekeeping": [
        "honey", "beekeeping"
    ],

    # Plantation / Horticulture
    "horticulture": [
        "orchard", "flowers", "lemongrass"
    ],

    # Bamboo & Forest Products
    "bamboo_forest": [
```

```
        "bamboo", "crafts", "silk", "art"
    ],

    # Organic / Compost
    "organic_products": [
        "vermicompost", "compost"
    ],

    # Value Added / Processed
    "processed_products": [
        "jaggery", "sweets", "wadi"
    ],

    # Farming / Services
    "services": [
        "farming", "hydroponics", "tailoring",
        "trainer", "seed", "production","paddy"
    ]
}

def classify_crop_category(text):
    text = text.lower()  # normalize
    for key, keywords in crop_categories.items():
        for word in keywords:
            if word in text:
                return key
    return "Others"

df['crop_category'] = df['crops'].apply(classify_crop_category)
```

In [38]:

```
counts = df['crop_category'].value_counts()

plt.figure(figsize=(7,7))
plt.pie(
    counts.values,                # sizes
    labels=counts.index,          # labels auto detected
    autopct='%1.1f%%'
)

plt.title("Categorical distribution on the basis of Crops")
plt.show()
df["state"].value_counts()
```



Out[38]:

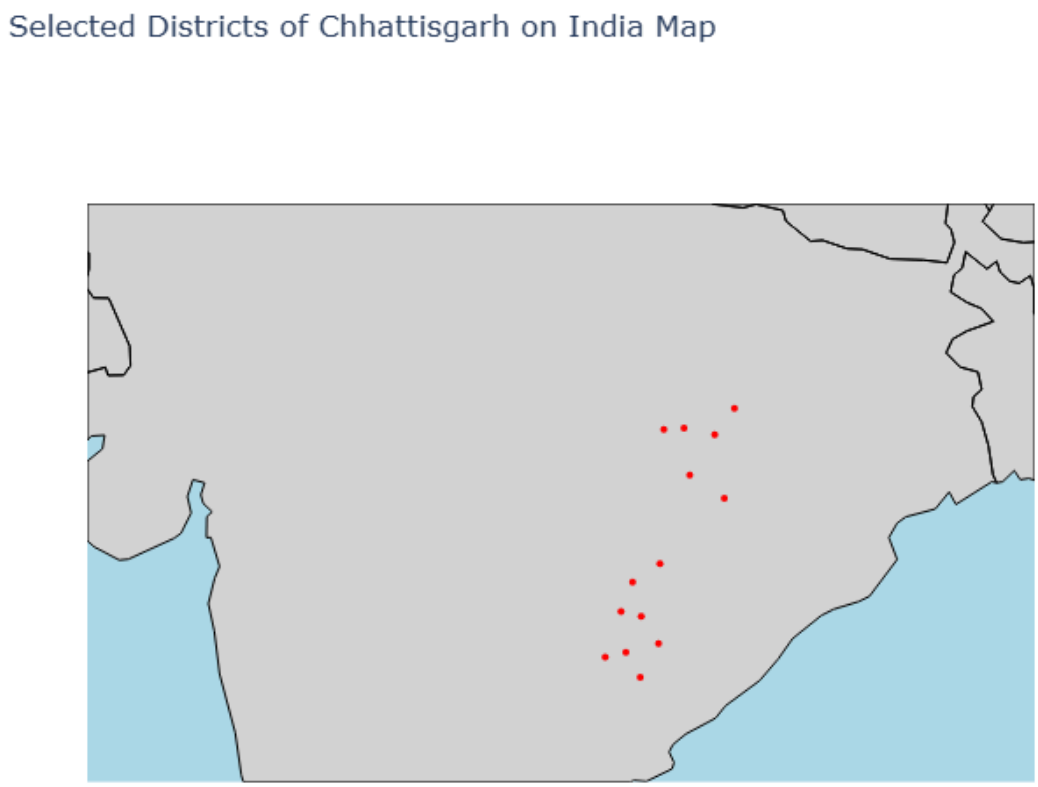
```
state
Maharashtra         6
Andhra Pradesh      5
Himachal Pradesh    5
Gujarat             5
Karnataka           4
Bihar               4
Chhattisgarh        3
Haryana             2
Punjab              2
Uttarakhand         2
Madhya Pradesh      2
Odisha              2
Jharkhand           2
Uttar Pradesh       2
Telengana           1
Rajasthan           1
Assam               1
Goa                 1
Tripura             1
West Bengal         1
Meghalaya           1
Name: count, dtype: int64
```

In [39]:
```python
import plotly.express as px
import plotly.io as pio
```

In [40]:
```python
# District data
data = {
    "District": [
        "Bastar (Jagdalpur)", "Dantewada", "Bijapur", "Narayanpur",
        "Kondagaon", "Sukma", "Kanker", "Surguja",
        "Balrampur", "Surajpur", "Koriya", "Korba",
        "Raigarh", "Gariyaband"
    ],
    "Latitude": [
        19.07, 18.90, 18.80, 19.70,
        19.60, 18.40, 20.27, 23.12,
        23.62, 23.22, 23.25, 22.35,
        21.90, 20.63
    ],
    "Longitude": [
        82.03, 81.35, 80.92, 81.25,
        81.67, 81.65, 81.49, 83.20,
        83.61, 82.14, 82.56, 82.68,
        83.40, 82.06
    ]
}

df1 = pd.DataFrame(data)

# Plot on India map
fig = px.scatter_geo(
    df1,
    lat="Latitude",
    lon="Longitude",
    hover_name="District",
    title="Selected Districts of Chhattisgarh on India Map"
)

# Set map to India
fig.update_geos(
    scope="asia",              # shows India clearly
    showcountries=True,
    countrycolor="black",
    showsubunits=True,
    subunitcolor="gray",
    showland=True,
    landcolor="lightgray",
    showocean=True,
    oceancolor="lightblue",
    center={"lat": 22, "lon": 80},   # center near India
    projection_scale=7               # zoom level
)

fig.update_traces(marker=dict(size=4, color="red"))
```

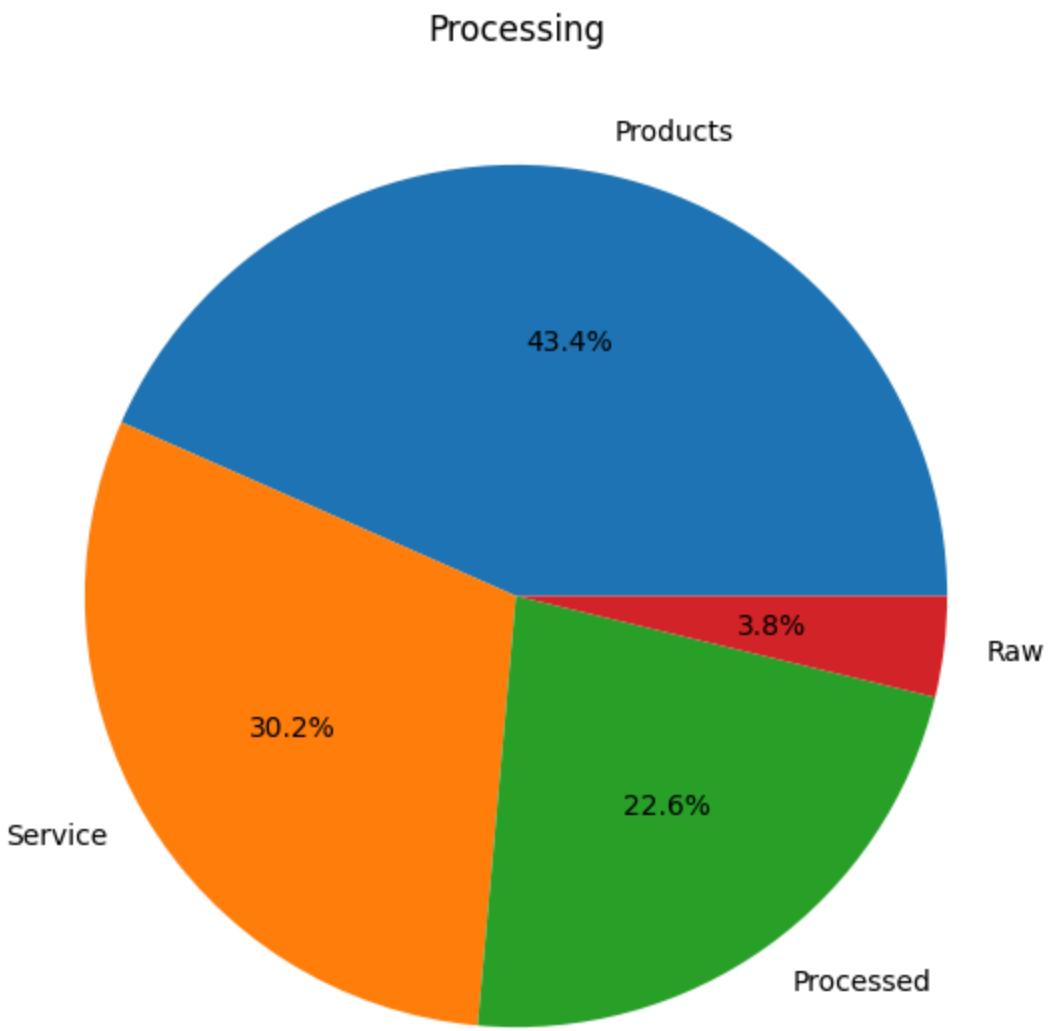
```
fig.update_layout(height=600)

fig.show()
fig.write_image("map.png")

# Display PNG in notebook
from IPython.display import Image
Image("map.png")
```

Out[40]:



In [41]:

```
counts = df['processing_type'].value_counts()

plt.figure(figsize=(7,7))
plt.pie(
    counts.values,                  # sizes
    labels=counts.index,            # labels auto detected
    autopct='%1.1f%%'
)

plt.title("Processing")
plt.show()
```

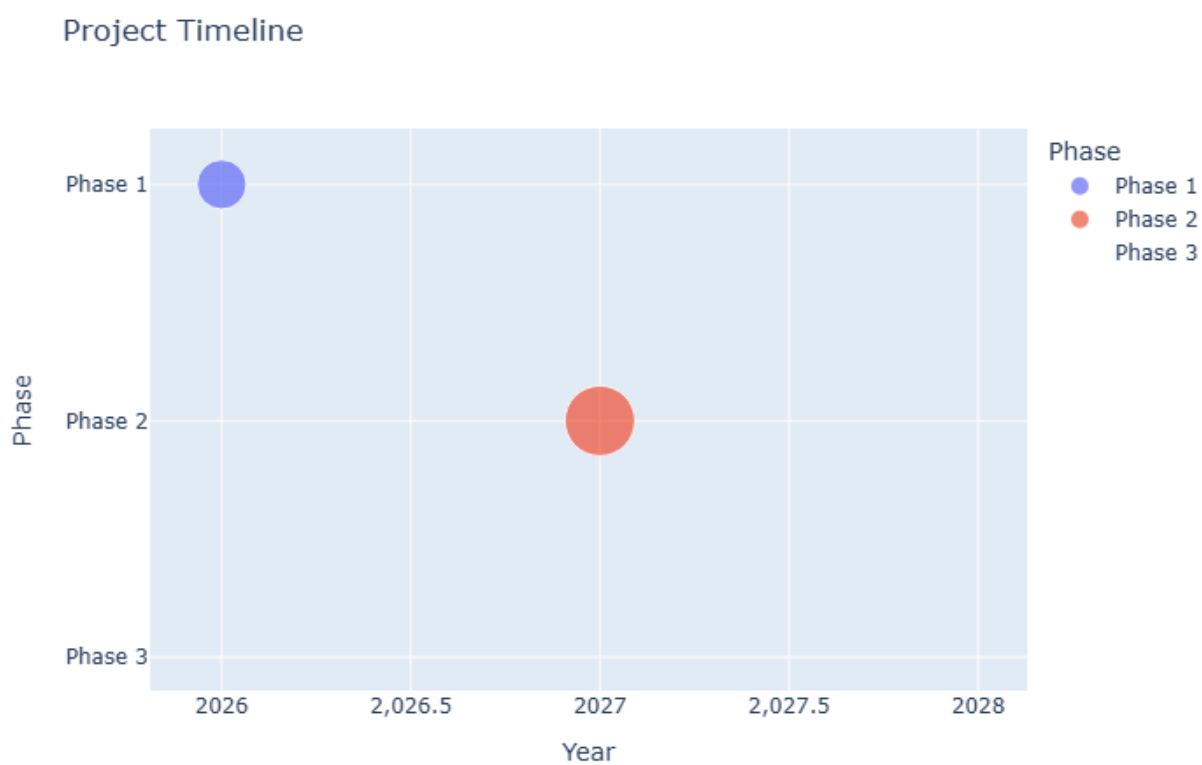


In [42]:

```
import pandas as pd
import plotly.express as px

# Timeline data
d = pd.DataFrame({
    "Phase": ["Phase 1", "Phase 2", "Phase 3"],
    "Year": [2026, 2027, 2028],
    "Funding (Cr)": [100, 200, 0],
    "Details": [
        "SHG Processing Hubs\n• Mushroom model\n• Mini dal mills",
        "FSSAI + Cold Storage\n• Mango/Cashew\n• Cafe franchises",
        "NGO Training Program"
    ]
})

# Timeline plot
fig = px.scatter(
    d,
```

```
    x="Year",
    y="Phase",
    size="Funding (Cr)",
    color="Phase",
    hover_data=["Details"],
    title="Project Timeline"
)

fig.update_traces(marker=dict(sizemode='area', sizeref=2.*max(d["Funding (Cr)"

fig.show()
fig.write_image("plot.png")

# Display PNG in notebook
from IPython.display import Image
Image("plot.png")
```

Out[42]:

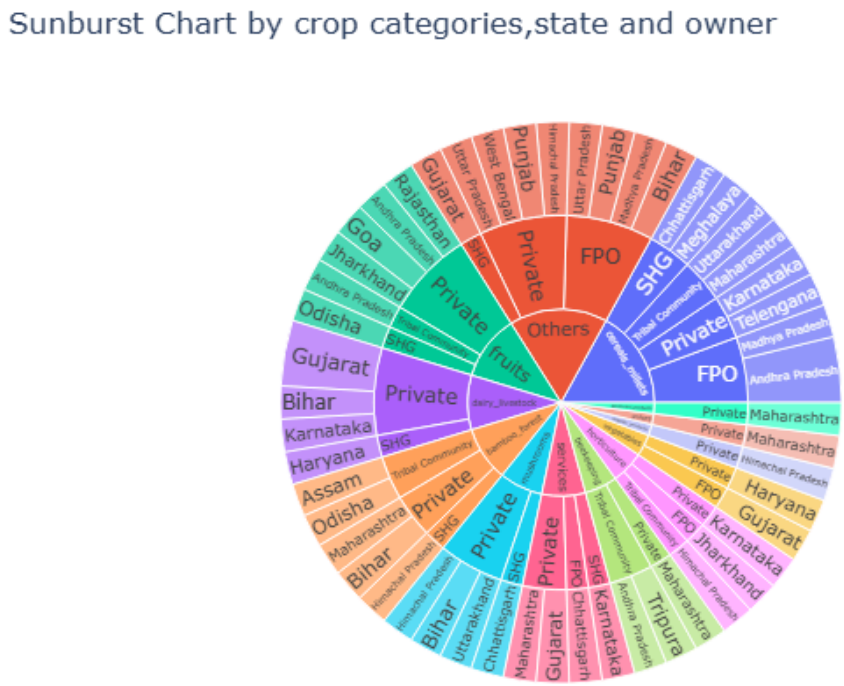


In [43]:

```
fig = px.sunburst(
    df,
    path=["crop_category", "owner", "state"],
    title="Sunburst Chart by crop categories,state and owner",
    width=700,
    height=500
)
fig.show()
fig.write_image("sun_burst.png")
```

```python
# Display PNG in notebook
from IPython.display import Image
Image("sun_burst.png")
```

Out[43]:



In [44]:

```python
df.to_excel("products_cleaned.xlsx", index=False)
```